



STANFORD UNIVERSITY • VERSION 0.1

SPHERE — *User Handbook*

Synthetic Privacy-preserving Honest Evaluation and Release Engine
The end-to-end platform for privacy-preserving synthetic data.

- 1 • GENERATE
- 2 • EVALUATE
- 3 • SHARE
- 4 • CATALOG
- 5 • ANALYSE

What is SPHERE?

The end-to-end platform for generating, certifying, and sharing privacy-preserving synthetic data.

SPHERE lets organisations produce a privacy-preserving synthetic twin of a sensitive dataset, formally evaluate and certify its quality, share it publicly, list it in a searchable catalog, and build rigorous analyses against it — all without the real data ever leaving the machine.

The five panels form a single pipeline. You do not have to complete every step — use as many as are relevant to your work.

Privacy Architecture

One rule governs all five panels: real data never leaves your machine.



CORE GUARANTEE

Your original data is only ever read from your local disk. It is never uploaded, transmitted, or copied to any external service — not SPHERE's servers, not Anthropic, not Dropbox, not Zenodo.

100% LOCAL — NEVER LEAVES YOUR COMPUTER

ITEM	WHY IT STAYS LOCAL
Your original (real) data	Never read beyond your local disk. No upload, no transmission.

ITEM	WHY IT STAYS LOCAL
Synthesis algorithm	Runs entirely in the app process on your machine.
Evaluation computation	All statistical comparisons (fidelity, privacy tests) run locally.
SPHERE certificate	Generated locally, saved as an HTML file on your disk.
API keys (Claude, Dropbox, Zenodo)	Stored encrypted via macOS Keychain. Never sent to SPHERE's servers.
Column names, dimensions & scores (AI auto-fill)	Derived from the synthetic file only — never the real one.
analysis.py and session figures (SPHERE AI)	Written and executed inside an isolated local sandbox only.

WHAT IS UPLOADED (SYNTHETIC DATA ONLY)

When you share a dataset, **only the synthetic CSV** is uploaded — never the real data.

- **Dropbox** — Synthetic CSV and certificate go to your account via OAuth. SPHERE has no access beyond the folder it creates.
- **Zenodo** — Synthetic CSV and certificate go to your Zenodo account via your personal API token.

WHAT THE CATALOG RECEIVES

When posting to SPHERE World, the API receives metadata only — no data rows, ever: title, description, domain, sharing terms, the public Dropbox/Zenodo link, aggregate scores, provenance flag, and certificate presence.

WHAT CLAUDE RECEIVES — CATALOG AUTO-FILL

The **Generate with AI** button in the Post panel sends: the synthetic CSV filename, column names (up to 10), row/column counts, and aggregate scores. **No cell values, no real data.**

WHAT CLAUDE RECEIVES — SPHERE AI PANEL

In the SPHERE AI tab Claude operates an agent loop over a local Python sandbox. It receives the **contents of the synthetic CSV** (via Python tools such as `pd.read_csv`), conversation messages, tool results (stdout/stderr from local execution), figures it requests to view, and files you explicitly attach.

What Claude never receives: the real CSV, its path, its contents, or any rows from it. The real data path is held in the app process only and is never included in any API message.

Deploy on real does not involve Claude. When you click that button, the *app* — not Claude — copies your real CSV into the sandbox, runs `analysis.py` as a local subprocess, then immediately deletes the copy. No Claude API call is made at this step. Claude's job was already finished when it wrote and verified the analysis on the synthetic data.

The SPHERE Workflow

1 Generate Synthetic Data

GENERATE TAB

Drag your real CSV into the app, or click to browse. The file is read locally — nothing is uploaded. Configure generation parameters (synthetic rows, noise level, random seed), then click **Generate**. The SPHERE algorithm runs on your machine and produces a synthetic CSV. Download it to a location of your choice.

Privacy: Your real data is only ever read from disk. It is never stored by the app or transmitted anywhere.

2 Evaluate & Certify

EVALUATE, CERTIFY & SHARE TAB

Load your real CSV and the synthetic CSV to evaluate. Click **Evaluate**. The app computes:

- **Fidelity scores** — mean, variance, correlations, and KS distance between real and synthetic.
- **Privacy scores** — resistance to singling-out, linkability, and inference attacks.

Click **Generate Certificate** to save a signed HTML evaluation report to your disk.

Privacy: All computation happens on your machine. No data leaves during this step.

Three options are available. In all cases only the **synthetic CSV** is uploaded — never the real data.

Option A — Dropbox

Connect via OAuth (directly with Dropbox, not SPHERE). Click **Share to Dropbox** — the synthetic CSV and certificate are uploaded and a public folder link is generated.

Option B — Zenodo

Enter your Zenodo API token (stored locally via Keychain). Click **Upload to Zenodo** — the files are uploaded and a permanent DOI is generated.

Option C — ZIP (local only)

Download a ZIP of the synthetic data and certificate for manual distribution. Nothing is uploaded.

Once the synthetic data is hosted on Dropbox or Zenodo, make it discoverable at **sphere-world.vercel.app**.

- 1 Click  **Post to SPHERE World** on your dataset card.
- 2 Fill in the title and description, or click **Generate with AI** to auto-fill using Claude (sends filename, column names, dimensions, scores — never cell values). Choose sharing terms:
 -  **Open (with citation)** — anyone can download; just cite your work.
 -  **Request access** — interested parties must email you first.
 -  **Revenue share** — commercial licensing terms apply.
- 3 Click **Post**. Only metadata and the public cloud link are sent — no data rows.
- 4 The button changes to  **Update on SPHERE World**. Click it any time to change the title, description, or sharing terms. The same catalog record is updated in place.

Deduplication: Each dataset has a stable local UUID. Re-posting always updates the same record. If the chosen title already exists in the catalog, you will be prompted to pick a different one.

Build a rigorous data analysis by chatting with Claude — using the synthetic data as a safe stand-in throughout development. When the analysis is ready, the *app* runs it locally against the real dataset. **Claude is not involved in that final execution step at all.**

Setup

- **Anthropic API key** — obtained at `console.anthropic.com/settings/keys`. Entered once, stored encrypted via Keychain, never transmitted to SPHERE's servers.
- **Model** — Claude Opus 4.7 (most capable), Sonnet 4.6 (balanced), Haiku 4.5 (fastest). Changed in the Settings panel; takes effect on the next message.

Session workflow

- 1 **Load a synthetic CSV.** Click the green *Synthetic* chip. An isolated sandbox folder is created and the synthetic file is placed inside it.
- 2 **(Optional) Load the real CSV.** Click the red *Real* chip. This path is stored in the app process only — never shared with Claude. It is needed only for the run-on-real step.
- 3 **Chat with Claude.** Type a prompt and press Enter (⇧Enter for new line). Built-in starter prompts:
 - "Summarize the dataset: column types, missing values, and key distributions."
 - "Identify the strongest associations between variables and quantify them."
 - "Propose three hypothesis-driven analyses, then run the most informative one."
- 4 **Claude writes and runs code.** The agent iteratively writes Python in `analysis.py`, installs packages (`matplotlib`, `seaborn`, `statsmodels` ...), and executes in the sandbox. Figures appear in the right panel. All execution is local.
- 5 **Iterate.** Ask follow-ups, change chart styles, add statistical tests. Click **Review analysis.py** to inspect the current script at any time.
- 6 **Run on real data.** Once the synthetic run shows a green exit code, click **Deploy on real**. The app:
 1. Copies your real CSV into the sandbox as a temporary file.
 2. Runs `analysis.py` locally as a subprocess against that copy.
 3. Immediately deletes the copy when the script finishes.
 4. Shows real-data figures alongside the synthetic figures.

No Claude API call is made during this step. Claude already finished when it wrote and verified `analysis.py` on the synthetic data. Deployment is a direct local execution by the app.

- 7 **Session report.** After a successful run, a self-contained HTML report is generated — full transcript, `analysis.py`, both figure sets, and an AI-written narrative. Click **Open report in sandbox** to reveal it in Finder.

File attachments

Click + in the chat input bar to attach files (PDFs, images, text/code). Attached files are sent to the Claude API as conversation context. Do not attach files containing real personal data.

Tools Claude can use

TOOL	WHAT IT DOES
<code>python</code>	Executes Python code in the sandbox. Output and errors are returned to Claude.
<code>pip_install</code>	Installs a Python package into the sandbox.
<code>read_file</code>	Reads a file from the sandbox directory.
<code>write_file</code>	Writes or overwrites a file in the sandbox.
<code>list_files</code>	Lists files in the sandbox directory.
<code>view_figure</code>	Renders a saved figure so Claude can inspect and refine it.

Buttons reference

BUTTON	WHAT IT DOES
Send	Sends your message to Claude. Press Enter (↵Enter for new line).
■ Stop	Interrupts Claude at the next turn boundary.
Review analysis.py	Shows the current analysis script in a dialog.
Deploy on real	App runs <code>analysis.py</code> locally against your real CSV. Claude is not contacted.
Sandbox folder	Opens the sandbox directory in Finder (scripts, figures, report).
↺ Reset	Clears the conversation and starts a fresh session with the same CSV.

First-launch note: The first time SPHERE AI is used, `matplotlib`, `seaborn`, and `statsmodels` are pre-installed in the sandbox (30–90 seconds). Subsequent sessions start immediately.

Managing Your Datasets

Delete: Click *Delete from Dropbox* (or Zenodo) on any card. The app removes the remote folder, the local history entry, and the catalog listing.

Sync: Click **Sync** in the SPHERE World tab. If a file was deleted outside the app, the sync removes the stale local entry and catalog listing.

Update a listing: Click  **Update on SPHERE World** on any card to change the title, description, domain, or sharing terms. The existing record is updated in place — no duplicate is created.

SPHERE vs. External Synthetic



SPHERE

Generated by the SPHERE algorithm in this app

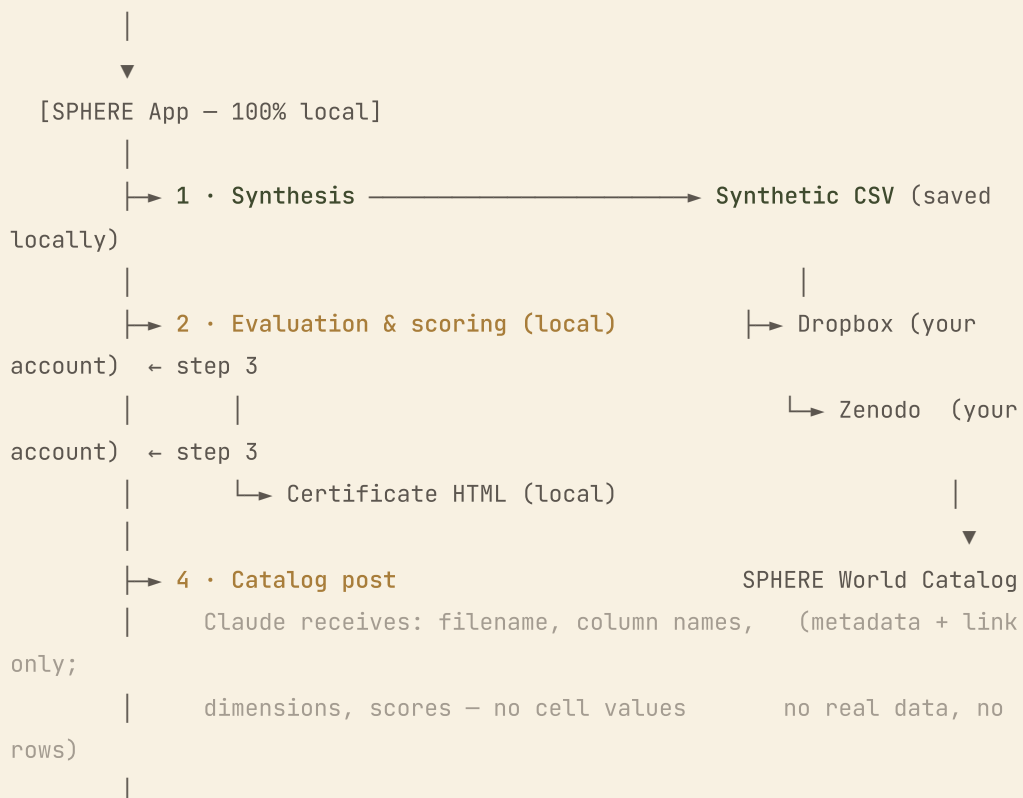


External synthetic

Generated by a third-party tool (SDV, synthpop, CTGAN...) and evaluated with SPHERE

Data Flow Summary

Your real data (CSV)



```
↳ 5 • SPHERE AI
    ↳ Synthetic CSV → sandbox (Claude sees this)
      Claude: python / pip / write_file / view_figure
      Claude receives: conversation + synthetic CSV contents
    ↳ Deploy on real ← app-only step — no Claude API call
      Real CSV copied → analysis.py runs → copy deleted
immediately
      Claude is not contacted. Real data never transmitted.
```

Frequently Asked Questions

Q: Does SPHERE ever see my real data?

No. Your real data is read locally by the app and never transmitted to SPHERE's servers, Anthropic, Dropbox, Zenodo, or any other service.

Q: Does Claude ever access my real data?

No — not in any step. In the SPHERE AI panel, Claude only ever sees the synthetic CSV and the code it writes itself. When you click "Deploy on real", the *app* copies your real CSV into the sandbox, runs `analysis.py` as a local subprocess, and immediately deletes the copy. No call is made to the Claude API during this step.

Q: Can the synthetic data be traced back to real individuals?

SPHERE evaluates exactly this risk using three attack models: *singling-out*, *linkability*, and *inference*. Privacy scores above 70–80 indicate strong resistance to each attack.

Q: Is it safe to use SPHERE AI with sensitive datasets?

Yes — provided the synthetic data has good privacy scores. Claude only ever receives the synthetic CSV, which by design does not contain real individuals' records. If the synthetic data has low privacy scores and residual real values, be aware that Claude can read those column contents via `pd.read_csv`.

Q: Can Claude access files outside the sandbox?

No. The Python sidecar runs under a macOS Seatbelt (`sandbox-exec`) profile that blocks all filesystem access outside the session sandbox directory. Any attempt returns a `PermissionError`.

Q: What exactly is sent to Anthropic's API?

Catalog auto-fill: filename, column names, dimensions, aggregate scores — no cell values.

SPHERE AI: conversation messages, tool results from local Python runs, figures Claude requests to view, files you explicitly attach, and the synthetic CSV's contents. Never the real CSV.

Q: Who can access the shared data on Dropbox or Zenodo?

That depends on the sharing terms you set in the catalog. "Open (with citation)" means anyone with the link can access the synthetic file. "Request access" means interested parties must email you first.

Q: What happens if I delete the dataset from Dropbox but it's in the catalog?

The next time you run **Sync**, the stale entry is detected and the catalog listing is removed automatically.

Q: Are my API keys and tokens stored securely?

All keys (Anthropic, Dropbox, Zenodo) are stored in your macOS user data directory, encrypted via macOS Keychain. They are never sent to SPHERE's servers.

Quick Reference

ACTION	WHERE	PRIVACY IMPACT
Generate synthetic data	Generate tab	100% local
Evaluate fidelity / privacy	Evaluate tab	100% local
Generate certificate	Evaluate tab	100% local
Upload to Dropbox / Zenodo	Share section	Synthetic data only
Post to SPHERE World catalog	SPHERE World tab → Post	Metadata + public link only
AI auto-fill description	Post panel → Generate with AI	Filename, column names, dimensions, scores → Claude API
Update catalog listing	SPHERE World tab → Update	Metadata only
Delete / Sync	Card → Delete · World tab → Sync	Removes from cloud storage + catalog
SPHERE AI — chat session	SPHERE AI tab	Synthetic CSV + conversation → Claude API
SPHERE AI — Deploy on real	SPHERE AI → Deploy on real	100% local app step – Claude not contacted
Session report	SPHERE AI → Open report	100% local – HTML in sandbox folder

